Exp NO : 4

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

Aim

To write a program to implement flow control at the Data Link layer using the Sliding window protocol, and to simulate the flow of frames from one node to another

Algorithm

1. Start the program
2. Read the total no. of frame to be sent (N), the sender's window size (W) & frame no. to simulate as lost during transmission.
3. Initialize base = 0 & nextSeq = 0
4. Repeat while base < N:
   a.) Determine the current Sender window as the range [base, base + w - 1]
   b.) Send all frame with sequence no. from nextSeq up to (base + w - 1), incrementing nextSeq after each frame, since the window restricts the sender to at most W outstanding frame.
   c.) If the lost frame lies with the frame just sent:
      i. The receiver discards the lost frame and any subsequent out-of-order frames.
      ii. The Sender times out and retransmits all frames starting from lost frame
   d) Else
      i. The receiver accepts each frame in order
      ii. On receiving the ACK for the last frame sent the Sender slide its window forward : base = nextSeq

5. Repeat step 4 until all N frames above been transmitted & acknowledge

6. Stop the program.

PROGRAM

```
def simulate(total_frame, window_size, lost_frame):
    base = 0
    next_seq = 0

while base < total frames:
    window_end = min(base + window_size - 1, total_frames - 1)
    print("\n ----------")
    print(f" Sender window : [ Frame {base} - Frame {window_end}")

    while next_seq < base + window_size and next_seq < total_frame:
        print(f" Node A --> Sending frame {next_seq}")
        if next_seq == lost_frame:
            print(f" (Frame {next_seq} LOST in Transit !)")
        next_seq += 1

    if lost_frame is not None and base <= lost_frame < next_seq:
        print(f"Node B --> Frame {lost_frame} missing. Discarding
            out-of-order frame after it.")
        print(f" Node A --> Timeout ! Resending window starting
            from frame {lost_frame}.")
        next_seq = lost_frame
        lost_frame = None
        continue

    for f in range(base, next_seq):
        print(f" Node B --> Received Frame {f}, Sending Ack {f}")
    print(f"Node A --> Ack {next_seq-1} received. Sliding
        window forward.")

    base = next_seq.
```

```
print ("\n - - - - - - - - - - - - ")
print (f" All {total_frames} frame Successfully transmitted
  from Node A to Node B.")

def main():
    print (" ===SLIDING WINDOW PROTOCOL ===\n")
    total_frame = int(input(" Enter tot no. of frames to Send : "))
    window_size = int(input(" Enter Sender's window size : "))
    lost_frame = int(input(f" Enter no. to Simulate as Lost in
      (0 to {total_frame-1}, enter -1 for no loss) : "))

    if lost_frame == -1:
        lost_frame = None

    Simulate(total_frames, window_size, lost_frame)

if __name__ == "__main__":
    main()
```

Result:

The program to implement flow control at the Data link Layer using the Sliding window protocol was successfully written in python, executed & verified by Simulating the flow of frame between two nodes. (Node A as receiver & Node B as Sender)

Output

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send : 8
Enter sender's window size : 3
Enter frame number to simulate as LOST (0 to 7, enter -1 for no loss) : 2

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Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
   (Frame 2 LOST in transit!)
Node B --> Frame 2 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 2.

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Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received, Sliding window forward.

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Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node A --> ACK 5 received, Sliding window forward.

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Sender Window : [Frame 6 - Frame 7]
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received, Sliding window forward.

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All 8 frames successfully transmitted from Node A to Node B.
```